Covers and Local Generation in Linear Storage Codes

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Abstract

We study cover structure and local generation as restrictions on linear storage codes. For the prescribed binary adjacency check on Clebsch, every connected binary-lift tower has kernel dimension at most $N/2 + 2$. Either equality and signing rigidity hold, or dimension is at most half and remains so at every descendant. The constant two is sharp at every level. The bound extends to regular deck groups $P \times D$ and $P \times S_3$, where P is a finite 2-group and D is odd abelian; it is sharp at every cyclic degree. Reciprocal width-r Clebsch checks have rank at least $6r$, equality exactly for flat gains, and no rank $6r + 1$. A width-five counterexample rules out a stronger general bound.

For homogeneous checks on a fixed free-group voltage template, under specified ambient pro-p and separating-chain hypotheses, disconnected matched ports in the infinite incidence cover imply asymptotic half rate at fixed width, uniformly over coefficient fields and values. For growing widths, excess rate requires positive dimension in simple coefficient factors exceeding every fixed size. Independent-chord covers with a spanning 2-factor have uniformly vanishing rate for nonzero scalar checks and for invertible blocks preserving a common full flag. Exact homogenization preserves rate but can create one large simple factor. Ordinary group-word recovery inherits the scalar cover bounds through its exponent check.

Separately, in characteristic two and with odd girth greater than $2R + 1$, the code dimension plus the dimension spanned by radius-R codewords is at most the sum of attained coordinate dimensions. A nonlinear encoder with recoverable affine hull consequently pays for excess support size or entropy through its nonlocal sources. Exact homogenization and the dimension interfaces are Lean-verified. The finite and asymptotic cover theorems and the graph-local and source/group-word applications have complete written proofs but are not formalized. No general nonlinear bound or native tape realization is supplied.

Contributions. Hans Brende chose the problem and directed the research. GPT-6 Astra (OpenAI) developed the arguments, text, and Lean formalizations under his direction. The imported rank-approximation theorem and other established inputs are attributed separately.

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1 Introduction

A recovery graph specifies which coordinates may determine another coordinate. It does not specify how a code is represented or computed. This distinction matters in the coding approach to matrix transposition: a structural property of a machine's dependency graph becomes useful only when a coding theorem applies to its actual code fibres. The companion paper [5] obtains long matched-port distances from native tape geometry and studies several coding bounds with additional premises. Here we examine two such premises more closely: algebraic structure in regular covers, and local generation of a linear code.

The cover results combine two settings. For a fixed template repeated over finite group quotients, homogeneous recovery has an asymptotic half-rate theorem uniform over coefficient fields and values at fixed width. We identify what its filtration argument needs when width grows. For the prescribed Clebsch adjacency check, exact finite bounds instead control binary towers and specified regular covers, with sharpness and rigidity and without a separating-chain hypothesis. For local generation the graph need not be a cover at all. Locally supported codewords are forced into the image of a check operator, while the whole code lies in its kernel. Rank-nullity gives a quantitative restriction on their span.

1.1 The results

The principal conclusions are the following.

(1) **Sharp finite Clebsch-cover bounds.** Every connected binary-lift tower above Clebsch has adjacency-code dimension at most $N/2 + 2$. Either equality and signing rigidity persist, or the dimension drops to at most half and stays there. The constant two is sharp at every level. We extend the bound to regular deck groups $P \times D$, for a 2-group P and odd abelian D, and to $P \times S_3$; every connected three-sheet cover has dimension at most 26. A named rigid three-sheet seed supports further binary towers. Reciprocal blocks have rank at least $6r$, with equality exactly for flat gains and no rank $6r + 1$. A concrete counterexample limits a stronger proposed extension (Section 9 and Appendices A and B).

(2) **Uniform half rate on fixed covers.** Under the precise ambient-group and chain hypotheses in Section 2, paired-port disconnection in the infinite incidence cover forces asymptotic rate at most $1/2$ for homogeneous linear checks of fixed total width. One threshold works for every coefficient tuple over every field of the fixed characteristic (Theorem 4.3). The coefficients and fields may therefore vary along the sequence.

(3) **An obstruction to excess rate at growing width.** Let $\alpha_i(k)$ be the fraction of internal dimension in simple coefficient-module factors larger than k, after algebraic closure. For each fixed k and $\varepsilon > 0$, eventually

$$r_i \leq (1 - \alpha_i(k))(1/2 + \varepsilon) + \alpha_i(k).$$

Thus a sustained excess δ over half rate requires at least 2δ of the dimension in factors exceeding every fixed cutoff (Theorem 5.1 and Corollary 5.2).

(4) **Uniform zero rate for a stricter check class.** On the independent-chord covers of a fixed graph with a spanning 2-factor, every homogeneous scalar check with nonzero edge weights has rate tending to zero. The convergence is uniform in the field, weights, and diagonal entries. It also holds for arbitrary-width blocks with a common full flag and invertible edge maps (Theorem 6.2 and Corollary 6.3).

(5) **The cost of removing sheet dependence.** Any sheet-dependent linear check has an exact homogeneous inflation with the same rate. Its internal width grows by the sheet count, and its coefficient algebra can be a full matrix algebra of that growing size. Hence this conversion does not supply the small-factor premise (Propositions 8.1 and 8.2).

(6) **A local-generation deficit.** In characteristic two and with no odd cycle of length at most $2R + 1$,

$$\dim \mathcal{C} + \dim \mathcal{L}_R(\mathcal{C}) \leq \sum_v \dim \pi_v(\mathcal{C}).$$

An above-half code must therefore have substantial dimension outside the span of its locally supported words (Theorem 10.2). If a nonlinear Cartesian-source encoder has a recoverable affine hull, nonlocal sources must pay for excess support size or entropy (Theorem 10.4). All-local sources generate a hull of at most half rate.

(7) **Group-word extensions of the cover bounds.** Ordinary-word recovery over finite p-groups reduces to the scalar exponent check modulo p. Homogeneity of that check transfers the fixed-width half bound, and nonzero homogeneous edge exponents transfer the independent-chord zero-rate bound. The alphabet groups may grow without affecting the threshold (Corollary 7.2).

1.2 Relation to existing work

Storage codes and index coding are established subjects. In particular, triangle-free graphs can support binary linear codes with rate approaching one [2, 1, 11]. We make no general half-rate claim for triangle-free graphs. Our local theorem instead constrains the span of local codewords on such graphs; recovery locality and generator locality are different conditions.

The cover argument uses standard free-group subgroup separability, presented here by finite Schreier-graph completion; see Stallings [9] for the classical graph method. The free-group algebra input is Cohn's Sylvester-domain theorem [7], as recorded in [8, Introduction]. The asymptotic input is precisely Jaikin-Zapirain and Souza's Theorem 8.1 [8]. We apply that theorem to the specified quotients, then prove a separate uniformization over coefficient fields and parameters. We do not claim a new general rank-approximation theorem. The parameter argument uses the finite-residue-field consequence of the weak Nullstellensatz [10].

The first paper formulates the tape problem and its coding program [4]; the subsequent local-barrier paper separates cheap encoding from cheap local recovery [6]. The fourth paper supplies depth reduction and the reduction from a correct trace to a storage-code fibre [3]. The directed-recovery companion proves native paired-distance bounds and an all-width matrix-label criterion on a different family [5]. Our cover theorems neither subsume that all-width criterion nor establish its hypotheses for arbitrary cones. They remove field and coefficient restrictions in a specified cover model and expose the remaining width obstruction.

Proof status and attribution. Exact homogenization and the dimension interfaces, including the quotient deficit and split-generator criterion, are Lean-verified. Their formal statements are authoritative; the English outlines are guides to them. The finite and asymptotic cover results, graph-local argument, and source and group-word applications have complete written proofs but are not formalized. The named rigid seed includes an explicit binary matrix-inverse certificate with a portable exact checker; that checker is not a Lean proof. In particular, the split-generator interface does not itself verify the graph parity and boundary argument. Every numbered result

states its status. We distinguish the applications developed here from their standard algebraic inputs; external priority for the resulting combinations has not been established.

1.3 Connection to transposition

Keep the series' notation: d is the matrix side length, $b(d) = \max\{1, \lceil \log_2 d \rceil\}$, and $L = d^2 b(d)$. Statement (T) asks for one fixed multitape machine transposing these inputs in $O(L)$ time. A coding lower-bound route needs a code deficit that exceeds a paid genie cost. The present theorems can supply such a deficit only after their representation and linearity hypotheses have been proved for the relevant fibres. They are also tests for constructions: an above-half candidate in these regimes must escape the stated module or local-generation restrictions. The missing native implications are listed in Section 11; (T) remains open.

2 Templates, quotients, and checks

Fix a prime p and a finitely generated abstract free group Γ. A finite directed voltage template consists of a finite nonempty set V of types and a finite set E of triples $e = (u, v, g)$ with $u, v \in V$ and $g \in \Gamma$. The triple permits a type-u input for a type-v receiver. Equal triples are collected before assigning coefficients. Words that represent different elements of Γ remain separate even if they later coincide in a quotient.

For a quotient $q : \Gamma \twoheadrightarrow Q$, the incidence lift I_Q has ports $L_{(v,x)}, R_{(v,x)}$ for $v \in V, x \in Q$, with edges

$$L_{(u,x)} \; -\!- \; R_{(v,q(g)x)} \qquad ((u, v, g) \in E). \tag{2.1}$$

This graph is undirected for connectivity and distance; the recovery relation itself need not be symmetric. The two ports of (v, x) are *matched*. The infinite lift I_Γ uses the identity quotient. Common right multiplication of sheets is a graph automorphism. Our separation hypothesis is

$$L_{(v,1)} \text{ and } R_{(v,1)} \text{ lie in different components of } I_\Gamma \quad \text{for every } v \in V. \tag{2.2}$$

It then holds at every sheet. The literal template is retained even when selected coefficients vanish or cancel.

Give type v a finite-dimensional space W_v over a field K, put $w_v = \dim_K W_v$ and $B = \sum_v w_v > 0$, and choose $C_e : W_u \to W_v$ for every $e = (u, v, g)$. Let T_e be its extension by zero to $W = \bigoplus_v W_v$. With $\rho_Q(g)e_x = e_{q(g)x}$, define the homogeneous check

$$M_Q = I - \sum_{e \in E} T_e \otimes \rho_Q(g_e), \qquad (M_Q x)_{v,y} = x_{v,y} - \sum_{e=(u,v,g)} C_e x_{u,q(g)^{-1}y}. \tag{2.3}$$

Its normalized nullity is $r_Q = \dim_K \ker M_Q / (B|Q|)$. Every linear subcode of this kernel has at most this rate. Over finite K, the kernel rate is $\log_{|K|} |\ker M_Q| / (B|Q|)$. Homogeneity means that C_e does not depend on the sheet y.

A linear storage space with deterministic recovery admits linear recovery maps: on the attained parent projection define the map taking a parent tuple to its determined coordinate, then extend it linearly by a basis. This observation does not force homogeneous choices when different sheets originally have different maps. Homogeneity is a separate hypothesis of the cover theorems.

The actual chain. For the rank limits, fix a faithful embedding of Γ in a free-by-$\mathbb{Z}_p$ pro-p group $\mathcal{G}$. This means that $\mathcal{G}$ has a closed normal free pro-p subgroup with quotient $\mathbb{Z}_p$. Fix descending normal open subgroups U_i with $\bigcap_i U_i = \{1\}$ in $\mathcal{G}$, and put

$$H_i = \Gamma \cap U_i, \qquad Q_i = \Gamma/H_i, \qquad q_i = |Q_i|. \tag{2.4}$$

The normalization is q_i, not $|\mathcal{G}/U_i|$ unless the two quotients coincide. No cofinality in the separate pro-p completion of Γ is assumed. The natural maps $Q_j \to Q_i$, $j \geq i$, are surjective and have p-group kernels.

The prime, template, embedding, and chain stay fixed in every cover limit below. Widths are fixed through Section 4; thereafter each growing-width statement specifies its hypotheses. Zero-width types may be omitted. When nonidentity within-type voltages occur, an early quotient may collapse one to 1; then the matrix is still a check but need not describe recovery without self-input. Separation excludes identity within-type voltages, and the finite template and trivial intersection ensure that no such collapse occurs at sufficiently late levels.

One filtration fact used throughout. If a linear map T preserves a finite filtration $0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m = X$, and T_j is its map on F_j/F_{j-1}, then

$$\dim \ker T \leq \sum_{j=1}^{m} \dim \ker T_j. \tag{2.5}$$

Indeed the map from $(\ker T \cap F_j)/(\ker T \cap F_{j-1})$ to F_j/F_{j-1} is injective and lands in $\ker T_j$. Sum dimensions. The filtration need not split, and quotient kernel vectors need not lift.

3 Finite separation and rank transfer

Lemma 3.1 (One finite separator)**.** Not formalized; a complete proof is provided.

Under (2.2), there is a finite quotient $q : \Gamma \twoheadrightarrow Q$ in which every matched port pair is still disconnected. This quotient need not be a p-group or a member of the chain (2.4).

Proof (not formalized). The finite base incidence multigraph has ports L_v, R_v and one edge per triple. Orient a traversal from L_u to R_v with label g and its reverse with label g^{-1}. The voltage of a path with chronological labels $a_1, \ldots, a_m$ is $a_m \cdots a_1$.

In a base component D, choose a root and a spanning tree. Let t_z be the voltage of the root-to-z tree path and let J_D be the subgroup of voltages of closed root walks. It is generated by the finitely many elements $t_y^{-1} a t_x$ from oriented edges $x \xrightarrow{a} y$: writing $a = t_y(t_y^{-1} a t_x)t_x^{-1}$ makes the tree labels telescope along a closed walk. The path voltages from x to y are exactly $t_y J_D t_x^{-1}$, by closing a path at the root, and conversely inserting a closed root walk.

If L_v, R_v are in different base components, their lifts are always disconnected. Otherwise their matched lifts in a quotient are connected exactly when

$$q(t_{R_v}^{-1} t_{L_v}) \in q(J_D).$$

Thus (2.2) gives a finite list of pairs (J, g) with J finitely generated and $g \notin J$.

Here is the required finite separation argument. In the coset graph with vertices zJ, retain the paths from J spelled by a finite generating set of J and by g. For a left action, traverse a word from its rightmost letter. Their finite vertex union S contains both J and the distinct vertex gJ. Each free generator acts on the retained edges by a partial injection of S into itself. Complete it to a

permutation, matching the unused domain and range vertices. Freeness extends these permutations to a homomorphism $\Gamma \to \mathrm{Sym}(S)$. The retained generator paths of J fix the vertex J, while g does not. The image of g is therefore outside the image of J.

Take the product of these finitely many separating homomorphisms and let Q be its image. Each forbidden membership fails in its designated factor, hence in Q. If the list is empty, use the trivial quotient. □

This is the standard subgroup-separation mechanism specialized to the two ports of every type. Its role is to supply a finite rank witness; it does not replace the actual chain by this auxiliary quotient.

Proposition 3.2 (A finite rank witness)**.** Not formalized; a complete proof is provided.
For any field K and widths of total $B > 0$, let $M = I - \sum_e T_e g_e \in \mathrm{Mat}_B(K[\Gamma])$ fit a template satisfying (2.2). Then

$$\mathrm{irk}_{K[\Gamma]} M \geq \lceil B/2 \rceil.$$

Here inner rank is the minimum k for a factorization through k columns over the group algebra.

Proof (not formalized). Use the quotient Q from Lemma 3.1. Order its incidence components uniformly at random. Select a coordinate (v, x) when its left component precedes its right component, and give it weight w_v. The components differ, so its selection probability is $1/2$. Some order selects total weight at least $B|Q|/2$.

For a selected input α and selected receiver β joined by a dependency, the left component of α equals the right component of β. Selection of β therefore places its left component strictly before that of α. The selected principal block of M_Q, ordered by left component, is triangular with identity diagonal. Consequently $\mathrm{rank}_K M_Q \geq B|Q|/2$.

A factorization $M = XY$ through k columns specializes under the regular representation to a factorization of M_Q through $k|Q|$ columns. Hence $k|Q| \geq B|Q|/2$ and $k \geq \lceil B/2 \rceil$. No semisimplicity is needed, even when p divides $|Q|$. □

Published algebraic inputs. Cohn's theorem identifies inner rank over a free-group algebra with rank in its universal division ring [7, 8]. For the subgroup and chain in (2.4), Jaikin-Zapirain–Souza [8, Theorem 8.1] gives, for each fixed matrix M over $\mathbb{F}_p[\Gamma]$,

$$\lim_i \frac{\mathrm{rank}_{\mathbb{F}_p} M_{Q_i}}{q_i} = \mathrm{irk}_{\mathbb{F}_p[\Gamma]} M. \tag{3.1}$$

These are imported theorems, not formalized here. The source uses a right action on rows. Its finite rank equals our left-action rank on columns: the pairing $\langle y, x \rangle = \tau(\sum_j y_j x_j)$, where τ extracts the identity coefficient in $\mathbb{F}_p[Q_i]$, is nondegenerate and satisfies $\langle y, Mx \rangle = \langle yM, x \rangle$.

Theorem 3.3 (Fixed prime-field checks)**.** Not formalized; a complete proof is provided.
With (2.4) and (2.2), every fixed $\mathbb{F}_p$ check of total width $B > 0$ satisfies

$$\lim_i \frac{\dim_{\mathbb{F}_p} \ker M_{Q_i}}{B q_i} \leq \frac{\lfloor B/2 \rfloor}{B}.$$

The same upper bound on the limsup holds when the $\mathbb{F}_p$ coefficients vary with i, while the template and widths remain fixed.

Proof (not formalized). Combine Proposition 3.2 with (3.1) and rank-nullity. There are finitely many $\mathbb{F}_p$ assignments to the fixed coefficient entries, so the maximum of their convergence thresholds works for every assignment at once. □

For a fixed finite extension $K/\mathbb{F}_p$ of degree a, restriction of scalars multiplies both kernel dimension and B by a. Thus the pointwise half bound holds over K, with the possibly stronger bound $\lfloor aB/2 \rfloor/(aB)$. This argument alone does not yield $\lfloor B/2 \rfloor/B$ over K or a uniform threshold over all a. The next section supplies the latter uniformity at the half constant.

Lemma 3.4 (Long paired distances imply infinite separation)**.** Not formalized; a complete proof is provided.

For a fixed template with $n = |V| > 0$, suppose finite quotient lifts satisfy, for every fixed integer $r \geq 0$,

$$\#\{(v, x) : \mathrm{dist}_{I_{Q_i}}(L_{(v,x)}, R_{(v,x)}) \leq r\} = o(nq_i).$$

Then (2.2) holds. This implication needs no nesting or prime-power condition on the quotients.

Proof (not formalized). An infinite-cover matched path at one type has finite length r. It descends to a walk of that length in every quotient. Right translation supplies such a walk at all q_i sheets of that type, giving proportion at least $1/n$, a contradiction. □

Thus the native paired-distance invariant can supply the separation premise *if* a fixed-template cover representation is already available. The rank proof provides no such representation.

4 Uniformity over coefficient fields

The pointwise rank limit does not by itself allow the field or coefficients to change at each level. We prove the required uniformity in two steps. This section first works with any group Γ and compatible finite quotients Q_i, provided every refinement kernel is a p-group.

Lemma 4.1 (p-refinement monotonicity)**.** Not formalized; a complete proof is provided.

Let K have characteristic p, and fix a square matrix $M \in \mathrm{Mat}_B(K[\Gamma])$ with $B > 0$. If $Q_j \twoheadrightarrow Q_i$ has kernel P, a finite p-group, then

$$\dim_K \ker M_{Q_j} \leq |P| \dim_K \ker M_{Q_i}, \qquad \frac{\dim_K \ker M_{Q_j}}{Bq_j} \leq \frac{\dim_K \ker M_{Q_i}}{Bq_i}.$$

The field and coefficients are fixed within this comparison.

Proof (not formalized). Choose a section $s : Q_i \to Q_j$. Each fine sheet is uniquely $s(x)k$ with $k \in P$, and left multiplication by g takes it to

$$s(g_i x)\kappa(g, x)k, \qquad \kappa(g, x) = s(g_i x)^{-1} g_j s(x) \in P.$$

If $M_{ab} = \sum_g c_{ab,g} g$, its fine block from coarse column (b, x) to coarse row (a, y) is the operator

$$\sum_{g : g_i x = y} c_{ab,g} \rho_P(\kappa(g, x)) \quad \text{on } K[P].$$

Every nonzero finite-dimensional $K[P]$-module has a nonzero fixed vector. Induct on $|P|$: a nontrivial finite p-group has a central element z of order p (by its class equation and Cauchy's theorem), and $(z - I)^p = 0$. Its nonzero kernel is P-invariant, with action factoring through $P/\langle z \rangle$;

induction applies there. For trivial P there is nothing to prove. Repeating on quotients gives a full flag in $K[P]$ with one-dimensional trivial factors.

Tensor this flag with the coarse coordinate space. On every quotient layer, each $\rho_P(\kappa)$ acts as 1, so the induced matrix is exactly M_{Q_i}. All colliding words contribute their scalar sum; none is discarded. Apply (2.5) to the $|P|$ layers and use $q_j = |P|q_i$. □

Theorem 4.2 (Finite-type uniformization)**.** Not formalized; a complete proof is provided.

Let $\mathscr{R}$ be a finite-type commutative unital $\mathbb{F}_p$-algebra, and let $M \in \mathrm{Mat}_B(\mathscr{R}[\Gamma])$, $B > 0$, have fixed finite support. Use compatible finite quotients with p-group refinement kernels. For a unital map $a : \mathscr{R} \to K$ to a field, write

$$r_i(a) = \frac{\dim_K \ker M_{Q_i}(a)}{Bq_i}.$$

Suppose $0 \le \beta < 1$ and $\limsup_i r_i(a) \le \beta$ for every such map to a finite *field. For every $\varepsilon > 0$ there is I such that*

$$r_i(a) < \beta + \varepsilon$$

for all $i \ge I$, all characteristic-p fields K, and all unital maps $a : \mathscr{R} \to K$.

Proof (not formalized). The zero ring has no unital maps to fields. If $\theta = \beta + \varepsilon > 1$, the assertion follows from $r_i \le 1$. Otherwise $0 < \theta \le 1$. Set $N_i = Bq_i$, $k_i = \lfloor (1-\theta)N_i \rfloor$, and let J_i be the ideal of $\mathscr{R}$ generated by all (k_i+1)-minors of M_{Q_i}. For every field specialization,

$$r_i(a) \ge \theta \quad \Longleftrightarrow \quad \operatorname{rank} M_{Q_i}(a) \le k_i \quad \Longleftrightarrow \quad a(J_i) = 0. \tag{4.1}$$

The rounding is exact, including $\theta = 1$.

If $J = \sum_i J_i$ were proper, choose a maximal ideal $\mathfrak{m} \supseteq J$. By the finite-type weak Nullstellensatz [10, Theorem 10.34.1], $\mathscr{R}/\mathfrak{m}$ is a finite extension of $\mathbb{F}_p$. Its residue map would be bad at every level by (4.1), contradicting the pointwise hypothesis. Thus $J = \mathscr{R}$ and there is a finite identity

$$1 = \sum_{\ell=1}^{m} f_\ell \Delta_\ell, \tag{4.2}$$

where Δ_ℓ is a generating minor at some level i_ℓ. Take $I = \max_\ell i_\ell$.

If a specialization were bad at $j \ge I$, then Lemma 4.1, applied to that same field and tuple, would make it bad at every i_ℓ. All $a(\Delta_\ell)$ would vanish, contradicting (4.2). This proves the claim. The ideals J_i need not be nested; only their bad field loci are ordered by monotonicity. □

This argument also allows equations among parameters and finitely many nonzero constraints, represented by inverse variables. It supplies an eventual threshold, not a useful numerical estimate for it. A search for (4.2) is possible when the ring, chain, and matrices are effectively presented and the threshold is rational; no running-time bound is asserted.

Theorem 4.3 (Field-uniform half rate)**.** Not formalized; a complete proof is provided.

Fix the template, embedding, and chain of Section 2, and suppose (2.2). Fix widths $w_v \ge 0$ of total $B > 0$. For every $\varepsilon > 0$ there is I such that for every $i \ge I$, every characteristic-p field K, and every homogeneous coefficient tuple fitting the template,

$$\frac{\dim_K \ker M_{Q_i}}{Bq_i} < \frac{1}{2} + \varepsilon.$$

The threshold is independent of K and the coefficient tuple.

Proof (not formalized). Give each scalar entry of each C_e an independent variable and form the universal check over their polynomial ring over $\mathbb{F}_p$. Every finite-field tuple has pointwise rate at most $1/2$ by Theorem 3.3 and restriction of scalars. Remove zero-width types if necessary; the remaining incidence graph retains separation. Apply Theorem 4.2 with $\beta = 1/2$. □

This rules out above-half homogeneous constructions with fixed width even when they change fields and coefficients at every size. It does not bound arbitrary sheet-dependent maps or nonlinear code supports. The p-kernel condition is material: in characteristic three, $1 + a$ has nullity zero on the trivial quotient but positive nullity on C_2. Thus Lemma 4.1 is false for unrestricted refinements.

5 Growing widths and coefficient modules

Keep the fixed geometry and separation of Theorem 4.3, but at stage i allow an arbitrary field K_i of characteristic p and arbitrary spaces $W_{v,i}$, of total dimension $B_i > 0$. The homogeneous coefficient maps may also vary. Choose an algebraic closure Ω_i of K_i and set $\overline{W}_i = (\bigoplus_v W_{v,i}) \otimes_{K_i} \Omega_i$. Let $\mathcal{A}_i$ be the unital Ω_i-algebra generated by the vertex projectors and the embedded coefficient maps T_e.

Take a composition series of the $\mathcal{A}_i$-module $\overline{W}_i$. Write its simple-factor dimensions as $d_{i,1}, \dots, d_{i,m_i}$. For $k \geq 1$ put

$$\alpha_i(k) = \frac{1}{B_i} \sum_{j : d_{i,j} > k} d_{i,j}. \tag{5.1}$$

These are dimensions over Ω_i, not over $\mathbb{F}_p$. All statements below hold for any composition series. The Jordan–Hölder theorem also makes this multiset independent of the chosen series; extension of a field to its algebraic closure does not change a finite matrix's rank, since rank is detected by minors.

Theorem 5.1 (A bound from simple-factor mass)**.** Not formalized; a complete proof is provided.
For each fixed $k \geq 1$ and $0 < \varepsilon < 1/2$, one index $I(k, \varepsilon)$ works for all fields, widths, and homogeneous coefficient tuples: for all $i \geq I(k, \varepsilon)$,

$$r_i \leq (1 - \alpha_i(k))(1/2 + \varepsilon) + \alpha_i(k).$$

Proof (not formalized). Every invariant subspace N is graded by types: $N = \bigoplus_v E_v N$, because the vertex projectors E_v are in $\mathcal{A}_i$, are orthogonal, and sum to the identity. The grading passes to every invariant subquotient, and each T_e still maps its source piece to its target piece. Thus the induced check on a simple factor fits the same template, with some nonnegative vertex dimension vector of total $d_{i,j}$.

Tensor the composition series with $\Omega_i[Q_i]$ and apply (2.5). For the factors of total dimension at most k, there are only finitely many possible vertex dimension vectors. Take the maximum of the corresponding thresholds from Theorem 4.3. This maximum is independent of the field, coefficients, and number of factors. Every such factor contributes at most $(1/2 + \varepsilon) d_{i,j} q_i$ to nullity. Each larger factor contributes at most $d_{i,j} q_i$. Sum, divide by $B_i q_i$, and use $\sum_j d_{i,j} = B_i$. Field extension preserves the original rate. □

Corollary 5.2 (Tightness and excess)**.** Not formalized; a complete proof is provided.
If

$$\lim_{k \to \infty} \limsup_i \alpha_i(k) = 0,$$

then $\limsup_i r_i \le 1/2$. *Conversely, if* $\liminf_i r_i \ge 1/2 + \delta$ *with* $0 < \delta \le 1/2$, *then for every fixed* k,

$$\liminf_i \alpha_i(k) \ge 2\delta.$$

A limsup excess gives only the corresponding limsup conclusion.

Proof (not formalized). For fixed k, take limsup in Theorem 5.1 and then let $\varepsilon \downarrow 0$. This gives $\limsup r_i \le 1/2 + (1/2)\limsup \alpha_i(k)$; now let $k \to \infty$. For the converse, rearrange the finite inequality as

$$\alpha_i(k) \ge \frac{r_i - 1/2 - \varepsilon}{1/2 - \varepsilon},$$

take liminf, and then let $\varepsilon \downarrow 0$. The same rearrangement with limsup proves the last assertion. No growing cutoff is substituted into a fixed-cutoff threshold. □

Corollary 5.3 (A common internal flag)**.** Not formalized; a complete proof is provided.

Suppose at each stage all $n = |V|$ *vertex spaces are identified with one space* Z_i *of arbitrary positive dimension, and the internal maps* C_e *preserve a common full flag after algebraic closure. Then* $\limsup_i r_i \le 1/2$, *uniformly in fields and dimensions. Pairwise commutativity of the identified internal maps suffices for the flag hypothesis.*

Proof (not formalized). The direct sums over types of successive internal flag spaces are invariant under every projector and embedded coefficient map. Their successive quotients have total dimension n. Refining them to composition series gives $\alpha_i(n) = 0$; apply Theorem 5.1.

For commuting operators over an algebraically closed field, an eigenspace of one operator is invariant under the others. Successively restricting to eigenspaces gives a common eigenvector. Its invariant line and induction on the quotient give a full common flag. This proves simultaneous triangularizability, not diagonalizability. The embedded maps between different types need not commute. □

Why factors are taken after algebraic closure. There is a strict difference from measuring simple factors over the prime field. Take two types with identity maps in both directions and one extra forward map given by multiplication by a generator ζ of $\mathbb{F}_{p^m}/\mathbb{F}_p$. Use voltages $1, 1, a$ over $\Gamma = \langle a \rangle$ with quotients C_{p^i} from $\mathcal{G} = \mathbb{Z}_p \times \mathbb{Z}_p$ and $U_i = p^i\mathbb{Z}_p \times p^i\mathbb{Z}_p$. Paired ports are disconnected because the two incidence type classes are disjoint. Over $\mathbb{F}_p$ the coefficient algebra is $\mathrm{Mat}_2(\mathbb{F}_{p^m})$: the identity maps give the cross matrix units and the extra map generates all field scalars in a diagonal corner. Its natural module is simple of dimension $2m$. Over an algebraic closure the separable field extension splits into m copies of that field, so the module has m factors of dimension 2. Hence $\alpha_i(2) = 0$ even when m grows without bound. This is a comparison of hypotheses; bipartiteness already bounds every code on this particular relation by half rate.

The excess conclusion is a construction requirement, not a claim that large simple factors produce high-rate codes. The explicit example in Section 8 has maximal large-factor mass and nevertheless obeys an elementary all-code half bound.

6 Nonzero checks on independent-chord covers

For a stricter check class the limiting rate is zero. Fix a finite simple connected graph H on n vertices with a spanning 2-factor, that is, disjoint cycles covering all vertices. Choose a spanning tree and orient its chords. Let Γ be the free group on these chords. Assign reciprocal voltages $g_{uv} = g_{vu}^{-1}$

to adjacent ordered pairs, with tree voltages 1 and one independent generator on each oriented chord. Here g_{uv} is in matrix row u, column v; it takes a sheet at type v to the sheet multiplied on the left by g_{uv} at type u.

Fix an embedding and chain as in (2.4) for this abstract Γ. Over a characteristic-p field K, choose arbitrary diagonal scalars d_u and nonzero scalars c_{uv} for every ordered edge. Define

$$T_{uu} = d_u, \qquad T_{uv} = c_{uv} g_{uv} \quad (uv \in E(H)), \qquad T_{uv} = 0 \quad \text{otherwise.} \tag{6.1}$$

Reverse coefficients need not agree or be inverses.

Lemma 6.1 (A surviving determinant monomial)**.** Not formalized; a complete proof is provided.

Abelianize the chord generators to independent Laurent variables $z_1, \dots, z_s$. The resulting matrix T_z has nonzero determinant over $K[z_1^{\pm 1}, \dots, z_s^{\pm 1}]$ for every allowed tuple.

Proof (not formalized). Orient the cycles of a spanning 2-factor and let π_0 send each vertex to its successor. The associated determinant term has nonzero coefficient $\operatorname{sgn}(\pi_0) \prod_u c_{u,\pi_0(u)}$. For any determinant permutation, record the net directed use of each undirected edge. This integer current has divergence zero; the Laurent exponents are its chord coordinates, with fixed choices of sign. If a second permutation has the same exponents as π_0, the difference of the two currents is supported on the spanning tree and has divergence zero. Leaf deletion shows that it vanishes.

Every oriented factor edge has net use one. Since a permutation uses each ordered edge at most once, that current forces use of the same oriented edge. These edges prescribe one outgoing arrow at every vertex, so the permutation equals π_0. The selected monomial is unique and cannot cancel, in any characteristic. No factor vertex uses a diagonal, so arbitrary d_u cause no issue. □

Theorem 6.2 (Uniform scalar rate zero)**.** Not formalized; a complete proof is provided.

For every $\varepsilon > 0$ there is $I(\varepsilon)$ such that for all $i \geq I(\varepsilon)$, all characteristic-p fields, all nonzero directed edge weights, and all diagonal values in (6.1),

$$\frac{\dim_K \ker T_{Q_i}}{nq_i} < \varepsilon.$$

Proof (not formalized). First fix one tuple over $K = \mathbb{F}_{p^a}$. Restrict scalars to $\mathbb{F}_p$ to obtain a fixed matrix T^{res} of size an over $\mathbb{F}_p[\Gamma]$. Its abelian specialization over $\mathbb{F}_p(z_1, \dots, z_s)$ is invertible. To justify this step, an $\mathbb{F}_p$-basis of K remains linearly independent over $\mathbb{F}_p(z)$: clear denominators and compare monomial coefficients. Its $\mathbb{F}_p(z)$-span is a finite-dimensional subalgebra of the field $K(z)$, hence is a field, since multiplication by any nonzero element is injective and therefore surjective. It contains K and $\mathbb{F}_p(z)$, so it is all of $K(z)$. The specialization is consequently the restriction of scalars of the invertible map T_z from Lemma 6.1.

Any factorization of T^{res} through fewer than an columns would specialize to such a factorization of an invertible matrix over this rational-function field. Hence its inner rank is an. Equation (3.1) gives a limiting normalized nullity of zero. Restriction of scalars preserves that normalized nullity, so the fixed finite-field tuple has rate tending to zero.

Now form the finite-type parameter algebra

$$\mathscr{R} = \mathbb{F}_p[c_{uv}, c_{uv}^{-1} : uv \text{ an ordered edge}][d_u : u \in V(H)].$$

Its maps to fields are exactly the permitted tuples. Apply Theorem 4.2 to the universal matrix with $\beta = 0$. The nonzero constraints survive every residue-field specialization. □

Corollary 6.3 (Arbitrary widths with a common flag)**.** Not formalized; a complete proof is provided.
Replace the scalars in (6.1) by internal matrices on one w_i-dimensional space at each type, over any characteristic-p field K_i. Assume that all edge and diagonal matrices preserve a common full flag after algebraic closure, and every edge matrix is invertible. Then the normalized nullity tends to zero uniformly in the fields, coefficients, flags, and widths $w_i \geq 1$.

Proof (not formalized). On each one-dimensional internal flag quotient, the edge matrices induce nonzero scalars: an invertible triangular matrix has nonzero diagonal entries. The induced diagonal scalars are unrestricted. There are w_i scalar check layers, each on nq_i coordinates. The same threshold in Theorem 6.2 bounds every layer's nullity by εnq_i. Equation (2.5) then bounds the full nullity by εnw_iq_i. Extending fields preserves rank, so the estimate holds over K_i. □

Commuting internal edge and diagonal maps give the flag by the proof of Corollary 5.3; edge invertibility is still required. For a recovery check, choose diagonal I and edge coefficients equal to the negatives of the recovery maps. Thus this theorem excludes the corresponding full-neighbor, common-flag construction class, even at unrestricted vector width. It is a bound on specified checks, not the storage capacity of the underlying graphs.

The nonzero premise cannot be dropped. Set every diagonal to 1, retain only one edge uv with coefficients -1 in both directions, and set all other edge coefficients to zero. Its two-type block is

$$\begin{pmatrix} I & -P \\ -P^{-1} & I \end{pmatrix}, \qquad P = \rho_{Q_i}(g_{uv}),$$

which has kernel dimension q_i, while all other types have identity checks. Its normalized nullity is $1/n$ at every level. No paired-distance hypothesis was needed for the rate-zero theorem; the independent chord variables and nonzero weights supplied the stronger rank certificate directly.

7 Ordinary group-word recovery

The scalar cover bounds also apply to a nonlinear class with arbitrarily large finite alphabets. An ordinary group word uses variables, their inverses, multiplication, and fixed constants. The expression at a coordinate is fixed on the whole code; arbitrary symbol maps and endomorphisms are not included here. The latter require the vector-layer benchmark of [5, Theorem B.5].

Lemma 7.1 (The scalar exponent check)**.** Not formalized; a complete proof is provided.
Let G be a finite p-group and $\mathcal{D} \subseteq G^N$ a set satisfying fixed ordinary-word equations $x_v = w_v(x)$. Let $E_{vu} \in \mathbb{F}_p$ be the signed exponent sum of variable x_u in w_v, and put $k = \dim_{\mathbb{F}_p} \ker(I - E)$. Then

$$|\mathcal{D}| \leq |G|^k.$$

For any law on a nonempty $\mathcal{D}$, its entropy is at most $k \log_2 |G|$.

Proof (not formalized). Use the terminating lower exponent-p central series $P_1 = G$, $P_{j+1} = P_j^p[P_j, G]$. Its termination and the central-layer argument are proved in [5, Theorem B.5]. Write $t_j = \dim_{\mathbb{F}_p}(P_j/P_{j+1})$, so $\sum_j t_j = \log_p |G|$. Fix an attained solution modulo P_j and choose representatives modulo P_{j+1}. Each further lift has the form $r_v a_v$, where a_v is in the central elementary abelian layer. Centrality makes the change in an ordinary word equal to $\sum_u E_{vu} a_u$: products add, inversion negates, and constants give zero. Thus the lift equations have linear part $(I - E) \otimes I_{t_j}$. A nonempty fibre is contained in an affine solution space of size p^{kt_j}. Multiplying the fibre bounds gives $|\mathcal{D}| \leq p^{k \sum_j t_j} = |G|^k$. Empty sets and trivial groups satisfy the cardinality bound directly. Entropy is at most the logarithm of the support size. □

This is the ordinary-word specialization of the classical central lifting method discussed in [5], not a new general group-equation theorem. In particular, a zero exponent sum is a statement about this word language, not a rule for erasing arbitrary nonlinear operations from a recovery circuit.

Corollary 7.2 (Uniform bounds for growing group alphabets)**.** Not formalized; a complete proof is provided.

Fix the template, prime, embedding, chain, and positive integer type widths of Theorem 4.3. At level i, use any nontrivial finite p-group G_i as the alphabet in each of the $N_i = Bq_i$ symbol positions. Let a nonempty code $\mathcal{D}_i \subseteq G_i^{N_i}$ have ordinary-word recovery from the permitted parents. Suppose its exponent matrix modulo p is homogeneous on the external sheets, with the block shapes of the fixed template. For every $\varepsilon > 0$, eventually

$$\frac{\log_{|G_i|} |\mathcal{D}_i|}{N_i} < \frac{1}{2} + \varepsilon,$$

with one threshold independent of G_i, its order, the words, and their constants.

On the independent-chord family of Theorem 6.2, use one G_i-symbol per vertex. If the exponent on each directed edge is homogeneous and nonzero modulo p, the same normalized rate tends to zero, uniformly in these choices. The corresponding entropy bounds hold for every law on each code.

Proof (not formalized). The exponent check $I - E_i$ is a characteristic-p homogeneous linear check with the stated fixed type widths. Apply Lemma 7.1 to bound the normalized logarithmic size by $\dim_{\mathbb{F}_p} \ker(I - E_i)/N_i$, then apply Theorem 4.3. For the second conclusion, the diagonal is 1 and the directed edge coefficients are the negatives of the nonzero exponent sums. They meet Theorem 6.2 exactly. Its uniform threshold gives the zero-rate conclusion. The entropy assertions follow from the final part of Lemma 7.1. □

The words themselves may vary between sheets: only their exponent matrices must be homogeneous. Reordering occurrences, changing constants, and inserting commutators can preserve this matrix. The number of group-symbol slots per type remains fixed, even though the group alphabet may grow. Arbitrary endomorphic expressions have matrix-valued central derivatives; the scalar argument above does not cover them. Nor does it supply a group-word representation or a charged native implementation for an arbitrary tape fibre.

This extends the cover obstruction beyond linear codes while retaining an explicit algebraic restriction on recovery. An above-half candidate in this cover model must violate the word or exponent hypotheses, or the cover hypotheses themselves. The rate-zero conclusion additionally requires every prescribed edge exponent to be nonzero; the single-edge example of Section 6 shows why that condition matters.

8 Exact homogenization and its width obstruction

It is natural to try to encode sheet-dependent coefficients inside larger homogeneous blocks. The following transformation is exact, but its module structure explains why it does not settle the unrestricted-width problem.

Proposition 8.1 (Rate-preserving inflation)**.** Lean: `sheetCheck`, `inflatedCheck`, `sheetReindex`, `reindex_inflated`, `kernelEquiv`, `finrank_kernel_inflated`, `rate_cross_product` [`HomogeneousInflation.lean`]

On a finite quotient Q of order q, allow coefficients indexed by the destination sheet:

$$(M_C x)_{v,t} = x_{v,t} - \sum_{e=(u,v,g)} C_e(t) x_{u,g^{-1}t}.$$

There is a homogeneous check on the same external voltage support, with each internal width multiplied by q, that is permutation equivalent to q copies of M_C. Its normalized nullity is unchanged.

Outline of the formal proof. The formalization takes $W = \bigoplus_v W_v$ and extends every $C_e(t)$ by zero from its source type to its target type. Thus one sheet is a vector in W, and the two checks below are exactly `sheetCheck` and `inflatedCheck` on W-valued functions. Replace W_v by $\widetilde{W}_v = \bigoplus_{t\in Q} W_v$ and set $(D_e z)_t = C_e(t) z_{g^{-1}t}$. The homogeneous check is $\widetilde{M} = I - \sum_e D_e \otimes \rho_Q(g_e)$. Its coordinates $x_{v,x,t}$ have external sheet x and internal sheet t, and a dependency simultaneously replaces these by $g^{-1}x, g^{-1}t$. Therefore $h = x^{-1}t$ is invariant. For each $h \in Q$ put $y^h_{v,t} = x_{v,th^{-1},t}$. This bijective reindexing turns the check into the original equations for y^h, independently for each h.

The formal kernel equivalence is $\ker \widetilde{M} \cong \prod_{h\in Q} \ker M_C$. Mathlib's dimension formulas therefore give $\dim \ker \widetilde{M} = q \dim \ker M_C$, while the ambient dimensions are respectively Bq^2 and Bq. The Lean rate identity is stated by cross multiplication, so it also holds at zero width; dividing by the positive ambient dimensions gives the stated rate equality. Neither commutativity of Q nor distinct voltages is assumed. □

For a linear subcode $\mathcal{C} \le \ker M_C$, the same coordinate permutation takes $\prod_{h\in Q} \mathcal{C}$ into the inflated kernel. Its dimension is $q \dim \mathcal{C}$, so it too preserves rate.

Proposition 8.2 (One large absolutely simple factor)**.** Not formalized; a complete proof is provided.
There is a fixed paired-separated template on an actual chain (2.4) for which the inflation in Proposition 8.1 produces a coefficient module that remains simple after every field extension, of dimension $2q_i \to \infty$. Nevertheless every storage code on the underlying two-type relation has rate at most one half for equal coordinate alphabets.

Proof (not formalized). Take $\Gamma = \langle a \rangle \subset \mathcal{G} = \mathbb{Z}_p \times \mathbb{Z}_p$ by $a \mapsto (1,0)$, with $U_i = p^i\mathbb{Z}_p \times p^i\mathbb{Z}_p$. Then $Q_i = C_{p^i}$. Use two types $0, 1$ and four triples

$$(0,1,1), \quad (1,0,1), \quad (0,1,a), \quad (0,1,a^2).$$

Their incidence edges stay in the disjoint classes $\{L_0, R_1\}$ and $\{L_1, R_0\}$, so every matched pair is disconnected in the infinite and finite lifts.

For scalar coordinates over $\mathbb{F}_p$, choose destination-sheet coefficients $1, 1, 1, \mathbf{1}_{t=1}$ in that order. On the inflated internal space $Z_i = \mathbb{F}_p^{Q_i}$, let $Pe_t = e_{at}$ and let F project onto the identity sheet. The four internal maps are I, I, P, FP^2. Include the two vertex projectors in their algebra. The cross identities put P and FP^2 in one diagonal corner. Since $P^{q_i} = I$, that corner also contains P^{-2} and therefore F. For any sheets x, y, powers of P give $P_x F P_y^{-1} = E_{xy}$, the corresponding matrix unit. The cross identities transfer these units to all four type corners. Consequently the algebra is $\mathrm{Mat}_{2q_i}(\mathbb{F}_p)$, and remains the full matrix algebra after any field extension. Its natural module is simple: a full endomorphism algebra can send any nonzero vector to any specified vector. Thus $\alpha_i(k) = 1$ eventually for every fixed k.

Finally each type is recoverable from the other type alone. Projection onto either full type is injective on any storage code. With q_i coordinates per type and common alphabet Σ, $|\mathcal{C}| \le |\Sigma|^{q_i}$. The same argument applies to the inflated block alphabet. No rate attainment by the displayed check is asserted. □

The example refutes automatic small-factor mass after homogenization. It does not refute a possible all-width half-rate theorem. In particular, the contrapositive of Corollary 5.2 provides a necessary algebraic escape condition, not a sufficient construction.

9 Finite Clebsch covers and rigidity

The preceding cover bounds are asymptotic and permit varying coefficients. Here we fix the ordinary adjacency check and obtain an exact finite bound without a separating-chain hypothesis. These are different restrictions; neither set of hypotheses is supplied by the other.

Let G be the Clebsch graph on $\mathbb{F}_2^4$, identified with the integers $0, \dots, 15$, where $u \sim v$ when $u \oplus v$ belongs to $\{1, 2, 4, 8, 15\}$. Write $M = I + A_G$. Equivalently, label the five generators by $i \in [5]$ and their ten pairwise sums by ij. The edges are 0–i, i–ij, and disjoint pairs. Over any characteristic-two field,

$$\operatorname{rank} M = 6, \qquad \dim \ker M = 10, \qquad M^2 = 0. \tag{9.1}$$

Indeed, pivoting the five independent singleton coordinates leaves a zero root row and column and the all-ones ten-pair block. For the last identity, the five Cayley translations commute and square to the identity, so $(I + \sum_i T_i)^2 = I + 5I = 0$. The binary code and its optimality among all storage codes on the base graph are treated in [3, Corollary H4], with Lean proofs there; that paper also attributes the original binary construction. The cover results below bound only the specified adjacency code and its subsets.

9.1 Exact loss under a binary lift

For a finite connected simple graph F, put $C_F = \ker(I + A_F)$ over $\mathbb{F}_2$. A signing $s : E(F) \to \mathbb{F}_2$ defines the lift with edges (u, b)–$(v, b + s_{uv})$. Let S_s be its symmetric zero-diagonal signed adjacency matrix and let

$$B_s(x, z) = x^T S_s z \quad (x, z \in C_F), \qquad q(s) = \operatorname{rank} B_s.$$

A signing is a *cut* if $s_{uv} = a_u + a_v$ for vertex values a_u. Call F *signing-rigid* if $B_s = 0$ implies that s is a cut. The reverse implication always holds.

Lemma 9.1 (Lift loss and rigidity propagation)**.** Not formalized; a complete proof is provided.
If $k = \dim C_F$, the binary lift F_s satisfies

$$\dim C_{F_s} = 2k - q(s).$$

The integer $q(s)$ is even. The lift is connected exactly when s is not a cut. If F is signing-rigid and $q(s) = 2$, then F_s is connected and signing-rigid.

Proof (not formalized). With $M_F = I + A_F$, the lifted check is $\left(\begin{smallmatrix} M_F + S_s & S_s \\ S_s & M_F + S_s \end{smallmatrix}\right)$. Writing a word as $(y + z, y)$ gives $M_F z = 0$ and $M_F y = S_s z$. Since $\operatorname{im} M_F = C_F^{\perp}$, the feasible z form $\operatorname{rad} B_s$; each has a k-dimensional fibre of choices of y. This proves the identity. The form is alternating. A nonzero alternating form splits off a nondegenerate two-dimensional summand; induction proves even rank.

For a cut, $S_s = D_a M_F + M_F D_a$, which vanishes as a form on C_F. Path integration gives the connectivity assertion: if all closed walks have sign sum zero, their path sums define a vertex potential; otherwise a closed walk joins the two sheets, and connectedness of F joins all vertices of the lift.

Suppose now that F is rigid, $q(s) = 2$, and a child signing t has zero form. Pairing diagonal child words $(x, x), (y, y)$ shows that the sum of its two signs above each base edge has zero base form, hence is a cut da. Switch the child vertices by a_v on sheet zero and by zero on sheet one. The two signs above each edge now agree, even on crossed edges, so the switched signing is the pullback of a base signing r. Switching preserves zero form.

Pair a diagonal word with a child word of sheet sum $z \in \operatorname{rad} B_s$. The resulting value is $B_r(x, z)$. Every such z occurs, by the lift equations. Thus B_r descends to the two-dimensional space $C_F/\operatorname{rad} B_s$. Over $\mathbb{F}_2$ its alternating form is either zero or the form induced by B_s. Rigidity makes r a cut or s plus a cut. Both pull back to cuts: the pullback of s is the cut given by the sheet-bit potential. Undoing the first switch proves child rigidity. Since $q(s) > 0$, s is not a cut, so the child is connected. □

Theorem 9.2 (C: sharp binary towers)**.** Not formalized; a complete proof is provided.

Start from Clebsch and take any sequence of connected binary lifts. At a level with N vertices, the binary adjacency code has dimension at most $N/2 + 2$. More precisely, either its dimension is $N/2 + 2$ and the graph is signing-rigid, or its dimension is at most $N/2$ and stays at most half at every descendant. Equality $N/2 + 2$ is attained at every level by repeatedly signing just one edge. The upper bound also holds for every connected regular cover of Clebsch with finite 2-group deck group, including nonabelian groups.

Proof (not formalized). Clebsch is signing-rigid by Corollary 9.4 below; that proof uses only block elimination, not the tower theorem. Write $e = \dim C_F - |V(F)|/2$. The lift identity gives $e' = 2e - q(s)$. Initially $e = 2$. At a rigid parent, a connected lift has positive even $q(s)$. If $q(s) = 2$, the new excess is two and rigidity propagates. If $q(s) \geq 4$, the new excess is nonpositive. Once nonpositive, it remains so without any rigidity assumption.

Endpoint evaluation of $\ker M$ onto $\mathbb{F}_2^2$ is surjective on every base edge uv. Column u of M has values $(1, 1)$ there; column w, for a different neighbor w of u, has values $(1, 0)$ by triangle freeness. Both are in the kernel by (9.1). Diagonal kernel words preserve this surjectivity in every binary lift. A single-edge signing therefore has $q(s) = 2$, proving sharpness and connectedness at every step.

A finite 2-group has a subgroup chain with successive indices two: use a central subgroup of order two and induct on the quotient. The intermediate quotient covers of a connected regular cover remain connected, and consecutive maps are binary lifts. Normality in the whole deck group is unnecessary. Apply the tower conclusion. □

9.2 Reciprocal checks and odd characters

Fix a characteristic-two field K and a positive block width r. A *reciprocal check* H has diagonal I_r, zero nonedges of Clebsch, and edge blocks $G_{uv} \in \mathrm{GL}_r(K)$ with $G_{vu} = G_{uv}^{-1}$. A vertex gauge replaces each block by $D_u^{-1} G_{uv} D_v$. The gains are *flat* if every based closed-walk product is the identity, equivalently if a vertex gauge makes every edge block I_r.

Theorem 9.3 (R: reciprocal rank rigidity)**.** Not formalized; a complete proof is provided.

Every such check has rank at least $6r$. Equality holds exactly for flat gains, and rank $6r + 1$ is impossible. In particular a scalar reciprocal check has rank at most seven exactly when it is diagonally similar to M; its rank is then six, and every other one has rank at least eight.

Proof (not formalized). The full block elimination, including the mixed root row and column, is given in Appendix A. It proves the lower bound, the equality criterion, and the forbidden next rank over every characteristic-two field. The scalar assertion is the case $r = 1$. □

Corollary 9.4 (The base signing rigidity)**.** Not formalized; a complete proof is provided.

The Clebsch graph is signing-rigid.

Proof (not formalized). A binary lift is a reciprocal width-two check, with edge blocks identity or swap. If $q(s) = 0$, Lemma 9.1 and (9.1) give nullity twenty and rank twelve. By Theorem 9.3 its gains are flat. Every closed walk therefore has sign sum zero, so path integration makes s a cut. This argument supplies the base rigidity algebraically, without a finite rank certificate. □

Theorem 9.5 (A: direct-product deck groups)**.** Not formalized; a complete proof is provided.

Let a connected regular Clebsch cover have deck group $P \times D$, where P is a finite 2-group and D is a finite odd abelian group. On its $N = 16|P||D|$ vertices the binary adjacency code has dimension at most $N/2 + 2$. In particular every finite abelian deck group is covered. The bound is sharp at every cyclic covering degree.

Proof (not formalized). Choose voltages (p_{uv}, d_{uv}), with inverse reverse voltages, and extend scalars to an algebraic closure of $\mathbb{F}_2$. This preserves rank. The regular representation of the odd abelian direct factor splits into its $|D|$ characters: commuting translations are diagonalizable, since their orders are odd, and character orthogonality gives each character once. The check therefore splits into weighted P-cover checks, with scalar edge weights $\chi(d_{uv})$.

Connectedness says that based closed-walk voltages generate $P \times D$. Every nontrivial character detects one of their D projections. Its weighted base check is reciprocal and nonflat, so its nullity is at most eight by Theorem 9.3. For any weighted check B, a binary lift with pulled-back weights has equations $Bz = 0$, $By = Sz$. The feasible z form a subspace of $\ker B$, and each fibre has dimension $\dim \ker B$. Thus nullity at most doubles. A subgroup chain of P bounds each nontrivial character sector by $8|P|$. For the trivial character the projected P cover is connected and unweighted, so Theorem 9.2 instead gives $8|P| + 2$. Summing the sectors gives $8|P||D| + 2$. The direct-product assumption is used in this decomposition; a semidirect product is not included.

For sharpness, put an m-cycle permutation T on one base edge uv, its inverse in reverse, and identity elsewhere. Deleting this edge leaves the base connected, so the cover is connected and regular cyclic. Put $M_m = M \otimes I_m$, let U embed the endpoint blocks, and let $V = U^T$. The endpoint surjectivity used in Theorem 9.2 implies that V is onto on $\ker M_m$. Symmetry then gives $\operatorname{im} U \cap \operatorname{im} M_m = 0$: if $Ua \in \operatorname{im} M_m$, its pairing with every kernel vector is zero, so surjectivity forces $a = 0$. The changed check is $M_m + U\Delta V$, where

$$\Delta = \begin{pmatrix} 0 & T - I \\ T^{-1} - I & 0 \end{pmatrix}.$$

Its kernel is exactly $\ker M_m \cap \ker(\Delta V)$, hence has dimension $10m - \operatorname{rank} \Delta = 10m - 2\operatorname{rank}(T - I) = 8m + 2$. More generally the last term is $8m + 2c(T)$, with $c(T)$ the number of permutation cycles. No diagonalizability of T is needed. □

The two bounds give rate at most $1/2 + 2/N$ for these prescribed checks. They rule out preserving Clebsch's 5/8 rate through these cover operations, even when no separating chain has been specified. They do not bound arbitrary fitting matrices or nonlinear codes on the lifted graphs. The tower dichotomy concerns binary towers; it is not asserted for all direct-product covers. Further small-representation consequences and a rigid three-sheet seed appear in Appendix B. No paired-port separation or native implementation follows from any of these constructions.

10 Local generation in characteristic two

This section is independent of the group-cover hypotheses. Let G be a finite simple undirected graph, let K be a characteristic-two field, let each W_v be finite-dimensional, and let $\mathcal{C} \leq \bigoplus_{v \in V(G)} W_v$ be

a linear code. For each v, a specified set $N_v \subseteq N_G(v)$ determines x_v on the code. The supplied recovery functions may be nonlinear; only the code support is assumed linear. Write $U_v = \pi_v(\mathcal{C})$ and $W = \bigoplus_v U_v$ for the attained coordinate spaces. Let

$$\mathcal{L}_R(\mathcal{C}) = \operatorname{span}_K\{x \in \mathcal{C} : \operatorname{supp} x \subseteq B_R(c) \text{ for some } c \in V(G)\}.$$

Here $B_R(c)$ is the closed graph-distance ball of radius R about c. The center may depend on the codeword. This is a condition on generators of the actual code, not on its recovery equations.

Lemma 10.1 (The dimension interface)**.** Lean: `kernel_range_dimension_bound`, `quotient_dimension_bound`, `split_generator_dimension_bound` [`LocalGeneration.lean`]

Let W be a finite-dimensional vector space over a division ring, $H : W \to W$ linear, and C, L subspaces with $C \subseteq \ker H$ and $L \subseteq \operatorname{im} H$. Then $\dim C + \dim L \le \dim W$. If also $L \subseteq C$, then

$$2 \dim C \le \dim(C/L) + \dim W.$$

For $H = I + A$ and $L = \operatorname{span} S$, the range premise follows whenever every $x \in S$ admits $x = s + As$ for some $s \in W$.

Outline of the formal proof. Monotonicity of dimension bounds the two summands by $\dim \ker H$ and $\dim \operatorname{im} H$. Mathlib's rank-nullity theorem says their sum is $\dim W$. Mathlib's quotient dimension formula gives the second inequality. For the last assertion, $x = (I + A)s$ puts each generator in the range; closure under linear combinations puts the span there as well. The paper library reuses these shared lemmas; no graph premise is hidden in their formal statements. □

Theorem 10.2 (The local span cannot exceed the deficiency)**.** Not formalized; a complete proof is provided.

If G has no odd cycle of length at most $2R + 1$, where $R \ge 0$ is an integer, then

$$\dim_K \mathcal{C} + \dim_K \mathcal{L}_R(\mathcal{C}) \le \sum_v \dim_K U_v. \tag{10.1}$$

Consequently local generation, $\mathcal{L}_R(\mathcal{C}) = \mathcal{C}$, implies the weighted half bound. More generally,

$$\dim_K(\mathcal{C}/\mathcal{L}_R(\mathcal{C})) \ge 2 \dim_K \mathcal{C} - \sum_v \dim_K U_v.$$

No degree, common-width, or reciprocal-recovery hypothesis is required.

Proof (not formalized). For each v, map an attained parent tuple to its determined coordinate in U_v. This map is well-defined by recovery and linear because both projections come from the linear space $\mathcal{C}$. Extend it from the attained parent subspace to $\bigoplus_{u \in N_v} U_u$ by extending a basis. We obtain fixed linear maps $A_{vu} : U_u \to U_v$ such that every codeword satisfies $x_v = \sum_{u \in N_v} A_{vu} x_u$. Let A be this ambient recovery operator on W, and put $H = I + A$. Characteristic two gives $\mathcal{C} \subseteq \ker H$.

Take a codeword x supported in $B_R(c)$. Color vertices of that ball by the parity of their distance from c. An edge within the ball whose endpoints had the same parity would, together with shortest paths to c, give an odd closed walk of length at most $2R + 1$. Removing repeated-vertex subwalks leaves an odd cycle no longer than that walk, contrary to the hypothesis. Thus every edge inside the ball joins opposite parities.

Split $x = s + t$ by these two parities, extending each part by zero outside the ball. We claim $As = t$ and $At = s$. At a vertex inside the ball, the only possibly nonzero contributions to its

recovery come from the opposite parity, so the assertion follows from the recovery equation for x. At a vertex v outside the ball, every neighbor u inside it must have distance exactly R from c:

$$R + 1 \le \operatorname{dist}(c, v) \le \operatorname{dist}(c, u) + 1 \le R + 1.$$

All these neighbors therefore have the same parity. One of the two sums is zero term by term; the other is the entire original recovery sum, which is zero because $x_v = 0$. This checks every boundary row, including its cancellations.

It follows that $Hs = s + As = s + t = x$. Every local codeword, and hence its span $\mathcal{L}_R(\mathcal{C})$, lies in $\operatorname{im} H$. Apply Lemma 10.1 to obtain (10.1). The quotient inequality is the second assertion of the same lemma. The split vectors need not themselves belong to the code, and we have not asserted $\operatorname{im} H \subseteq \mathcal{C}$. □

The dimension, quotient, and split-generator interfaces are formalized. The application to attained coordinate spaces, graph parity, and boundary cancellation above remains a written proof. At $R = 0$, a one-vertex-supported recoverable word is zero, so $\mathcal{L}_0(\mathcal{C}) = 0$, as the proof also shows. Zero-dimensional attained spaces and disconnected graphs cause no exception. Disjoint edges with equal values at their endpoints attain the half bound with radius-one generators. Conversely the triangle's parity code has dimension two on three scalar vertices and is generated inside radius-one balls; it violates half rate when the odd-cycle exclusion is removed.

Corollary 10.3 (Above-half codes need nonlocal dimension)**.** Not formalized; a complete proof is provided.

Put $D = \sum_v \dim_K U_v$. Under the hypotheses of Theorem 10.2, if $\dim_K \mathcal{C} \ge (1/2 + \delta)D$, then $\dim_K(\mathcal{C}/\mathcal{L}_R(\mathcal{C})) \ge 2\delta D$. In particular, for scalar characteristic-two codes on triangle-free graphs with N vertices and rate $1 - o(1)$, $\dim_K \mathcal{L}_1(\mathcal{C}) = o(N)$.

Proof (not formalized). Substitute the assumed dimension in the quotient inequality. For the scalar conclusion use $D \le N$ and $\dim \mathcal{L}_1(\mathcal{C}) \le D - \dim \mathcal{C} \le N - \dim \mathcal{C} = o(N)$. □

Thus the known near-unit-rate triangle-free binary families [1, 11] necessarily have a negligible span of radius-one codewords. This consequence does not contradict local recovery: many global codewords can satisfy local equations without being sums of local codewords. For a construction, the theorem identifies a required nonlocal part of the code. For an impossibility argument, it leaves the obligation to prove local generation, together with the appropriate odd-cycle condition, for the actual code under study. Bounded causal depth alone supplies neither assertion.

10.1 Nonlinear sources and a recoverable affine hull

Local generation can be supplied by local changes to a nonlinear encoder. The additional requirement is recovery on the encoder's entire affine hull. Recovery on its image alone does not suffice.

Theorem 10.4 (Payment for exceptional sources)**.** Not formalized; a complete proof is provided.

Retain the graph, parent sets, and radius of Theorem 10.2, and suppose K is finite, with $q = |K|$. Let I be a finite set, let each A_i be finite and nonempty, and let

$$F : \prod_{i \in I} A_i \longrightarrow \bigoplus_v W_v, \qquad S = \operatorname{im} F, \qquad \operatorname{aff}_K S = a + \mathcal{C}.$$

Assume that each specified parent set recovers its coordinate on all of $a+\mathcal{C}$. Fix a set $B \subseteq I$ of exceptional sources. For each $j \notin B$, assume there is a center c_j such that changing only source j changes no output coordinate outside $B_R(c_j)$. Put

$$k = \dim_K \mathcal{C}, \quad r = \sum_v \dim_K \pi_v(\mathcal{C}), \quad \mathcal{L} = \mathcal{L}_R(\mathcal{C}), \quad \ell = \dim_K \mathcal{L}, \quad M = \prod_{i \in B} |A_i|.$$

Each fixed exceptional-source fibre has image in one affine coset of $\mathcal{L}$. Consequently,

$$2 \log_q |S| \le r + \log_q M. \tag{10.2}$$

For any probability law of a source tuple Z on the stated Cartesian domain, with $X = F(Z)$ and entropies in bits,

$$2H(X) \le r \log_2 q + H(Z_B). \tag{10.3}$$

Neither independence nor full support of this law is required. If $B = \varnothing$, then $\mathcal{C} = \mathcal{L}$ and $2k \le r$. If $S = a + \mathcal{C}$, then also

$$q^{\dim_K(\mathcal{C}/\mathcal{L})} \le M;$$

if, in addition, X is uniform on this affine space, then $\dim_K(\mathcal{C}/\mathcal{L}) \log_2 q \le H(Z_B)$.

Proof (not formalized). Translation by a transfers recovery from $a+\mathcal{C}$ to $\mathcal{C}$. Thus Theorem 10.2 gives $k + \ell \le r$. Two source assignments differing only in a nonexceptional coordinate have output difference in $\mathcal{C}$, supported in its specified radius-R ball, and hence in $\mathcal{L}$. Any two assignments with the same exceptional tuple can be joined by changing the remaining coordinates one at a time. All intermediate assignments are in the Cartesian domain. Summing the differences proves that their outputs lie in one affine $\mathcal{L}$-coset.

There are at most M such fibres, each with at most q^ℓ outputs, whereas $S \subseteq a + \mathcal{C}$ gives $|S| \le q^k$. Adding the logarithms of these two bounds and using $k + \ell \le r$ proves (10.2). For any source law the same support bounds give

$$H(X) \le k \log_2 q, \qquad H(X \mid Z_B) \le \ell \log_2 q.$$

Combine the second with $H(X) \le H(Z_B) + H(X \mid Z_B)$, add the first, and apply $k + \ell \le r$ to obtain (10.3).

If B is empty, choose $a \in S$. The coset assertion puts $S - a$ in $\mathcal{L}$. Since $S - a$ spans $\mathcal{C}$, this proves $\mathcal{C} = \mathcal{L}$. Finally, the quotient of $F(Z) - a$ modulo $\mathcal{L}$ is a function of Z_B. If $S = a + \mathcal{C}$, that function maps the exceptional-source domain onto $\mathcal{C}/\mathcal{L}$, proving the quotient cardinality bound. A uniform X induces the uniform law on this quotient, whose entropy is $(k - \ell) \log_2 q$. A function of Z_B has entropy at most $H(Z_B)$, proving the last assertion. □

The quantity r measures affine coordinate dimensions. The theorem does not replace it by $\sum_v H(X_v)/\log_2 q$ or by $\sum_v \log_q |\pi_v S|$. For uniform X on the entire affine hull, these quantities do agree with r. Empty source sets, singleton alphabets, constant encoders, and zero-dimensional coordinates are included. At radius zero, $\mathcal{L} = 0$, so nonexceptional sources cannot affect the output.

The all-local conclusion recovers the weighted half bound for locally generated linear codes: choose finitely many ball-supported generators, use their coefficients as the sources, and let F be their sum. The more general statement permits nonlinear source dependence and charges the actual exceptional tuple. To apply it to a native fibre, one must supply the Cartesian representation, the local influence sets, and recovery on the whole affine hull. A small circuit or a shallow recovery cone supplies none of these automatically.

Example 10.5 (Recovery can fail on the affine hull)**.** On the bipartite graph $K_{3,3}$, the binary encoder

$$(a, b, c) \longmapsto (a, b, c,\ c + ab,\ a + c + ab,\ b + c + ab)$$

has eight outputs and exact neighbor recovery. Writing the last three coordinates as y_1, y_2, y_3, the inverse formulas are $a = y_1 + y_2$, $b = y_1 + y_3$, and $c = y_1 + ab$. Each source affects only the radius-one star centered at its own left vertex. Nevertheless, the affine hull has dimension four, with basis

$$(1, 0, 0, 0, 1, 0), \quad (0, 1, 0, 0, 0, 1), \quad (0, 0, 1, 1, 1, 1), \quad (0, 0, 0, 1, 1, 1).$$

The first three vectors are outputs at the three unit sources; the fourth is $F(1, 1, 0) + F(1, 0, 0) + F(0, 1, 0)$. They span every output and are independent, as their first three coordinates show. Their span contains $(0, 0, 1, 0, 0, 0)$, so the third coordinate is not recoverable there from its neighbors. All six coordinate dimensions are one; the inequality $2 \dim \mathcal{C} \leq 6$ would fail if recovery on the affine hull were omitted. The original eight-word code still has exactly half rate.

11 Construction requirements and the next questions

The results constrain different objects. The asymptotic cover theorems concern the coefficient algebra of a homogeneous check. The finite Clebsch theorems concern a prescribed adjacency check and the rigidity of its restricted signing forms. They do not require the asymptotic theorems' separating chain, but do not control all checks on the same graphs. The local theorem concerns the span of localized words in the code itself. The theorems do not supply either premise from the other. Their common use is to separate a promising construction from a family whose capacity is already controlled.

Hypothesis required of a candidate	Consequence here	What an above-half candidate must change
Clebsch adjacency check on the specified binary towers or regular covers	Dimension at most $N/2 + 2$, sharp in the stated families	The check or cover class to sustain a fixed positive excess in rate.
Fixed separated template, specified pro-p chain, homogeneous fixed widths	Half rate over every characteristic-p field	The representation, homogeneity, fixed-width condition, or code class.
Same template and chain, growing widths with tight simple-factor mass	Half rate; excess δ forces at least 2δ large-factor mass	A positive fraction of dimension in arbitrarily large absolute simple factors.
Independent-chord check, nonzero scalar edges, or invertible common-flag blocks	Rate tends to zero	Edge nonvanishing, the common flag, or the specified check family.
Characteristic two, odd girth greater than $2R + 1$, radius-R local generation	Weighted half rate	A quantitatively nonlocal code subspace, or the graph premise.
Same graph, Cartesian sources with recoverable affine hull	$2H(X) \leq r \log_2 q + H(Z_B)$	Sufficient exceptional-source entropy, or one of the representation premises.

Table 1: These are conditional tests, not automatic properties of tape computations.

11.1 A focused constructive target

The first target is a fixed separated voltage template and an actual chain of the stated type, with homogeneous coefficients whose geometric simple-factor dimensions grow. An above-half sequence would have to meet the mass lower bound of Corollary 5.2. Large factors alone are not evidence of high rate: the exact inflation control in Proposition 8.2 has them and still has a separate all-code half bound. A candidate should therefore be tested both for large factors and for an independent coding obstruction.

If the search instead uses characteristic-two codes on graphs of large odd girth, Corollary 10.3 gives a second test. Local recovery equations must leave enough independent global codewords outside the local span. A short description of the equations or a sparse check matrix is insufficient; the needed object is the actual kernel or subcode, with its dimension proved. The nonlinear-source extension offers a more concrete test: changing a local source must produce an actual local word of a recoverable affine hull. Recovery only on the encoder image cannot replace hull recovery. For group-word constructions on the cover families, increasing the alphabet size alone does not evade the bounds: homogeneous scalar exponents still give the uniform consequences of Corollary 7.2.

11.2 The corresponding impossibility targets

Question 11.1 (Arbitrary Clebsch covers)**.** *Does every finite connected permutation cover of Clebsch have binary adjacency-code dimension at most $N/2 + 2$?*

The finite results answer this for the specified towers and deck groups, not for every monodromy action. Any general argument must retain more than reciprocal block structure: Proposition A.3 refutes the proposed bound $\operatorname{rank} H \geq 8r - h - h^*$ for that larger class. Even an affirmative cover answer would concern the prescribed check, not unrestricted storage codes or native transposition.

Question 11.2 (Unrestricted block width)**.** *Under the fixed template, separation, embedding, and chain hypotheses of Theorem 4.3, does half rate hold for every sequence of homogeneous characteristic-p checks, without a condition on their simple-factor dimensions?*

An affirmative answer would, by Proposition 8.1, also control arbitrary sheet-dependent linear checks on those same quotient relations. Conversely, such a sheet-dependent above-half check would inflate to a homogeneous counterexample with growing width. This is an equivalence within the specified cover model; it does not extend the template or group hypotheses.

Question 11.3 (A native source of the needed structure)**.** *Can actual shallow-cone fibres of a putative linear-time transposer be placed, with a fully charged cost, in one of the code classes controlled here or in an all-alphabet extension of those classes?*

The native paired-distance theorem of [5] alone does not give a fixed voltage template, homogeneous recovery, linearity of the fibre, an internal flag, or locally supported generators. These are the main missing implications. Even a successful coding step must be combined with a compatible cheap depth reduction and, for adaptive machines, a paid description of the input-dependent trace or relation.

A Reciprocal Clebsch blocks: proofs and limits

This appendix proves Theorem 9.3 and records the exact identities that explain both the bound and its limits. All vector spaces here are over a characteristic-two field K. The width is $r \geq 1$.

A section is a family x_u satisfying $G_{uv}x_v = x_u$ on every edge. A dual section is a family of rows satisfying $\lambda_u G_{uv} = \lambda_v$. Write h, h^* for their dimensions. Spanning-tree propagation identifies them with the common fixed vectors and covectors of based holonomies; each dimension is at most r. Either dimension equals r exactly when the gains are flat.

A.1 The Schur complement and the first rank gap

Gauge the five root edges to identity. Write $X_{i,p} = H_{i,p}$ when $i \in p$, and pivot the singleton block I_{5r}. The resulting root/pair Schur matrix S satisfies

$$\operatorname{rank} H = 5r + \operatorname{rank} S, \qquad S_{00} = 0, \qquad S_{pp} = I_r,$$
$$S_{0,ij} = X_{i,ij} + X_{j,ij}, \qquad S_{ij,0} = X_{i,ij}^{-1} + X_{j,ij}^{-1},$$

and its ten-pair block B has

$$B_{pq} = \begin{cases} X_{i,p}^{-1} X_{i,q}, & p \cap q = \{i\}, \\ G_{pq}, & p \cap q = \varnothing, \end{cases} \qquad B_{pp} = I_r.$$

In particular all off-diagonal pair blocks are invertible and reciprocal.

Lemma A.1 (The minimum pair-block identity)**.** Not formalized; a complete proof is provided.
If $\operatorname{rank} B = r$*, then* $\operatorname{rank} H = 8r - h - h^*$.

Proof (not formalized). Fix a pair p_0. Pivoting $B_{p_0 p_0} = I_r$ shows $B_{pq} = B_{pp_0} B_{p_0 q}$. Put $T_p = B_{p_0 p}$ and gauge pair p by T_p^{-1}. All pair blocks become I_r. The shared-index identities imply $X_{i,p} = C_i$, independently of p containing i. All original disjoint-pair gains are identity. Define $A_{ij} = C_i + C_j$ and $b_{ij} = C_i^{-1} + C_j^{-1}$. Then

$$S = \begin{pmatrix} 0 & A \\ b & J_{10} \otimes I_r \end{pmatrix}.$$

The disjoint-pair graph is connected: two intersecting pairs have their complementary pair as a common neighbor. Consequently sections are specified by a common pair vector v for which all $C_i v$ coincide; dual sections by a row λ for which all λC_i coincide.

Pivoting pair p_0 leaves

$$\begin{pmatrix} D & R \\ T & 0 \end{pmatrix}, \quad D = A_{p_0} b_{p_0}, \quad R = [A_p + A_{p_0}]_{p \neq p_0}, \quad T = \left[b_p + b_{p_0} \right]_{p \neq p_0},$$

where R is horizontal and T vertical block stacking. Comparing overlapping pairs shows that a row annihilates R exactly when all its λC_i coincide; hence $\operatorname{rank} R = r - h^*$. Likewise $w \in \ker T$ exactly when all $C_i^{-1} w$ coincide, so $\operatorname{rank} T = r - h$. For such w, $b_{p_0} w = 0$, hence $Dw = 0$. The displayed matrix therefore has kernel $\ker T \times \ker R$, and rank $\operatorname{rank} R + \operatorname{rank} T$. Including the pivots proves the identity. □

Proof of Theorem 9.3 (not formalized). The singleton pivot and one pair diagonal block give $\operatorname{rank} H \geq 6r$. At equality, $\operatorname{rank} B = r$, and Lemma A.1 gives $h + h^* = 2r$. Thus both equal r and the gains are flat. Conversely flat gains gauge to $M \otimes I_r$, which has rank $6r$.

Suppose $\operatorname{rank} H = 6r + 1$. Every three-pair principal block of B has rank at most $r + 1$. After block vertex gauges it has the form

$$\begin{pmatrix} I & I & I \\ I & I & X \\ I & X^{-1} & I \end{pmatrix},$$

whose rank, by pivoting the first block, is $r + 2\operatorname{rank}(X - I)$. Thus $X = I$. Every triangle of the complete pair-block gain system is flat, which gives $\operatorname{rank} B = r$ by an anchor-pair gauge. Now Lemma A.1 implies $\operatorname{rank} H - 6r = (r - h) + (r - h^*)$. Nonflat gains make both summands positive; flat gains make both zero. Neither case permits rank $6r + 1$. □

A.2 Two-edge sharpness and a genuine counterexample

Proposition A.2 (Two-edge identity)**.** Not formalized; a complete proof is provided.

Put arbitrary $X, Y \in \mathrm{GL}_r(K)$ on the Cayley edges 0–1 and 0–2, with inverse reverse gains and identity elsewhere. Then $\operatorname{rank} H = 8r - h - h^*$. For transitive permutation gains this gives nullity $8r + 2$ on the connected r-sheet cover.

Proof (not formalized). The identity-edge complement is connected, using the detours 0–4–5–1 and 0–4–6–2. Thus h, h^* are the common fixed-vector and fixed-covector dimensions of X, Y. Evaluation of $\ker(M \otimes I_r)$ at $0, 1, 2$ is onto: the columns of M at $4, 5, 6$ restrict there to the three unit vectors and lie in the kernel. With U the three-block embedding and $V = U^T$, symmetry again gives $\operatorname{im} U \cap \operatorname{im}(M \otimes I_r) = 0$. The actual update is $U\Delta V$, where

$$\Delta = \begin{pmatrix} 0 & X - I & Y - I \\ X^{-1} - I & 0 & 0 \\ Y^{-1} - I & 0 & 0 \end{pmatrix}.$$

Its horizontal part has rank $r - h^*$, and its vertical part rank $r - h$, with disjoint domains and outputs. The updated kernel has dimension $10r - \operatorname{rank} \Delta$, proving the claim. In a transitive permutation action fixed vectors and covectors are constant on the single orbit, so $h = h^* = 1$. □

The two dimensions cannot be identified for arbitrary blocks. For $X = I + E_{12}$ and $Y = I + E_{13}$ at width three, they are $h = 1, h^* = 2$; Proposition A.2 gives rank 21, disproving a putative lower bound $8r - 2h = 22$.

Proposition A.3 (Failure of the general dual-balanced bound)**.** Not formalized; a complete proof is provided.

The inequality $\operatorname{rank} H \geq 8r - h - h^*$ fails for general reciprocal Clebsch blocks, even commuting determinant-one involutions. At width five there is a check with rank 34, $h = 1$, and $h^* = 4$.

Proof (not formalized). On the four Cayley square edges $e_1 = 01, e_2 = 13, e_3 = 23, e_4 = 02$ put $G_{e_j} = I_5 + E_{1,j+1}$, with one-based matrix indices. Put identity elsewhere. The nilpotent parts have all pairwise products zero, so the gains commute, are involutions, and have determinant one.

Let $C = \ker M$. The evaluations $\ell_0, \dots, \ell_3$ on C are independent, witnessed by columns $4, 5, 6, 7$ of M. For $e = uv$, the restricted edge form $B_e = \ell_u \wedge \ell_v$ has image $\langle \ell_u, \ell_v \rangle$. The concatenation of the four selected forms therefore has rank four. Ordering by coordinate layers, the check has diagonal blocks M and top row $[M, S_{e_1}, S_{e_2}, S_{e_3}, S_{e_4}]$, with all other off-diagonal blocks zero. The four lower vectors lie in C. Their upper equation is soluble exactly when $\sum_j B_{e_j} y_j = 0$, because $\operatorname{im} M = C^\perp$. The lower tuples have dimension $40 - 4 = 36$; each has a ten-dimensional upper fibre. Hence nullity is 46 and rank $80 - 46 = 34$.

The identity-edge complement is connected: replace every selected u–v by u–$(u \oplus 4)$–$(v \oplus 4)$–v. Sections and dual sections are consequently constant on this complement. The four gains fix exactly $\langle f_1 \rangle$ among vectors and the annihilator of f_1 among covectors. Thus $h = 1, h^* = 4$, while $8r - h - h^* = 35 > 34$. □

These gains are not permutation matrices. The counterexample therefore does not refute the $N/2+2$ bound for arbitrary connected permutation covers, which remains open here. It does rule out deriving that bound from the proposed unrestricted reciprocal-block inequality. The exact two-edge and minimum pair-block identities retain their stated premises.

B Small representations and a rigid three-sheet seed

Theorem B.1 (A fixed section at symplectic width two)**.** Not formalized; a complete proof is provided. *For a reciprocal Clebsch check with edge gains in* $\mathrm{SL}_2(K)$*, where* K *has characteristic two, rank below sixteen forces a nonzero section. Its only possible ranks below sixteen are twelve and fourteen; rank twelve is equivalent to flatness, and nonflat rank fourteen occurs.*

Proof (not formalized). Put $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Since $G^T J G = J$ for $G \in \mathrm{SL}_2(K)$, the form $\operatorname{diag}(J)H$ is alternating. The singleton Schur pivot of Appendix A gives $\operatorname{rank} H = 10 + \operatorname{rank} S$, where $\operatorname{diag}(J)S$ is alternating. If $\operatorname{rank} H < 16$, its radical quotient is a nondegenerate alternating space E of dimension two or four. The coordinate maps f_u into E satisfy $\langle f_u x, f_v y\rangle = x^T J S_{uv} y$. Each pair image $U_p = \operatorname{im} f_p$ is a nondegenerate plane. Two such planes intersect: otherwise they span the four-dimensional E, but their frame Gram matrix is

$$\operatorname{diag}(J, J)\begin{pmatrix} I & B_{pq} \\ B_{pq}^{-1} & I \end{pmatrix},$$

of rank two, contradicting nondegeneracy.

A pairwise-intersecting family of planes either has a common line or lies in a three-space. To see this, choose distinct U, V, put $L = U \cap V$ and $W = U + V$. A plane avoiding L meets U, V in distinct lines and lies in W. A plane outside W could then meet U, V only in L, making intersection with the plane avoiding L impossible. If all planes coincide, the common-line case applies.

In that case choose $0 \neq \ell \in \bigcap_p U_p$ and define $f_p v_p = \ell$. Comparing its pairings with f_p gives $B_{pq} v_q = v_p$. The shared-index identities make $X_{i,p} v_p = w_i$ independent of p containing i. Pairing ℓ with the root gives $\langle f_0 x, \ell\rangle = x^T J(w_i + w_j)$, independent of ij. Comparing overlapping pairs makes all w_i equal to one nonzero w. The values v_p at pairs and w at root and singletons form the required section, including every disjoint-pair edge.

In the three-space case the restricted alternating form on the span of the pair planes has rank two, so $\operatorname{rank} B = 2$. The pair gauge in Lemma A.1 can be chosen in $\mathrm{SL}_2(K)$ and makes all $B_{pq} = I$. We then have $X_{i,p} = C_i \in \mathrm{SL}_2(K)$ and $A_{ij} = C_i + C_j$. Pivoting one pair in the alternating Schur form leaves $\left(\begin{smallmatrix} D & JR \\ R^T J & 0 \end{smallmatrix}\right)$ of rank at most two, where $R = [A_p + A_{p_0}]$. For any Q, the rank of $\left(\begin{smallmatrix} D & Q \\ Q^T & 0 \end{smallmatrix}\right)$ is at least $2 \operatorname{rank} Q$: a kernel vector has first component in $\ker Q^T$, and each second-component fibre is a translate of $\ker Q$. Thus $\operatorname{rank} R \leq 1$. Choose a nonzero row λ annihilating R. Overlapping-pair comparisons give $\lambda C_i = \mu$ for the same nonzero row μ. The matrices $F_i = C_1^{-1} C_i$ fix μ and have determinant one. In a basis with $\mu = (1, 0)$ they have form $\left(\begin{smallmatrix} 1 & 0 \\ a_i & 1 \end{smallmatrix}\right)$, so fix $\ker \mu$ pointwise. For $0 \neq v \in \ker \mu$, all $C_i v = C_1 v = w$ coincide. Use v at pairs and w at the root and singletons, and undo the gauges.

The form is alternating, so its rank is even. The lower bound and flatness criterion in Theorem 9.3 leave only twelve and fourteen below sixteen. A single transvection $I + E_{12}$ on one edge adds rank at most two to the flat rank-twelve check and gives nontrivial holonomy, so its rank is fourteen. □

Corollary B.2 (Three sheets and a nonabelian odd action)**.** Not formalized; a complete proof is provided.

Every connected three-sheet Clebsch cover, including nonregular covers, has binary adjacency-code dimension at most 26*. Every connected regular cover with deck group $P \times S_3$, for a finite 2-group P, has dimension at most $N/2 + 2$, where $N = 96|P|$. The bounds* 26 *and* 50 *are attained for three sheets and regular S_3 sheets, respectively.*

Proof (not formalized). The permutation module $\mathbb{F}_2^3$ splits as its constant line and even-sum plane. The latter is the natural two-dimensional module for $S_3 = \mathrm{SL}_2(\mathbb{F}_2)$, acting on its three nonzero vectors. Transitive monodromy fixes no nonzero natural vector. By Theorem B.1 its check has rank at least sixteen and nullity at most sixteen. The constant sector contributes ten, giving the three-sheet bound.

Write $S_3 = \langle c, t : c^3 = t^2 = 1,\ tct = c^{-1} \rangle$. In $\mathbb{F}_2 S_3$, the central idempotent $e = 1 + c + c^2$ has ideal with basis e, et, isomorphic to $\mathbb{F}_2 C_2$. The complementary ideal has dimension four. In the natural representation take

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here e maps to zero, and I, C, T, CT are linearly independent. Thus the complementary ideal is $M_2(\mathbb{F}_2)$, with two natural column modules under left multiplication. The regular $P \times S_3$ check splits accordingly into an ordinary connected $P \times C_2$ quotient, contributing at most $16|P| + 2$ by Theorem 9.2, and two natural weighted P checks. Their base holonomy is full S_3, so each has base nullity at most sixteen. Binary-lift doubling, valid also for block weights, bounds each by $16|P|$. The sum is $48|P| + 2$. No semisimplicity of the $\mathbb{F}_2 C_2$ sector is assumed.

For sharpness put $U = I + E_{12}$ on the Cayley edge 0–1, $V = I + E_{21}$ on 0–2, and identity elsewhere. The identity-edge complement is connected by the detours in Proposition A.2. These gains generate S_3, so both actions give connected covers. The natural check has rank at most $12 + 2 + 2 = 16$, and at least sixteen by Theorem B.1. The binary quotient signs the two selected edges. Its restricted form has rank at most two, since its signing matrix is $e_0(e_1 + e_2)^T + (e_1 + e_2)e_0^T$. A cycle containing one selected edge shows the signing is not a cut; base rigidity forces rank two. Thus its nullity is eighteen. The three-sheet and regular checks have nullities $10 + 16 = 26$ and $18 + 2 \cdot 16 = 50$, respectively. □

B.1 A named rigid seed and its descendants

Let F be the three-sheet graph just constructed, with sheets the nonzero vectors $(1,0), (0,1), (1,1)$ and vertex encoding $3u + a$ for $a \in \{0, 1, 2\}$. The selected edge permutations are $(0, 2, 1)$ and $(2, 1, 0)$. It has 48 vertices and 120 edges.

Lemma B.3 (An exact finite signing certificate)**.** Not formalized; a complete proof is provided.

The graph F is signing-rigid.

Proof (not formalized). Its nullity is 26 by Corollary B.2 (also by Proposition A.2). In a basis $c_0, \ldots, c_{25}$ of its kernel, the signing map has entries

$$Q_{(i,j),uv} = c_i(u)c_j(v) + c_i(v)c_j(u), \qquad i < j.$$

Order edges and pairs lexicographically. The accompanying exact certificate `tools/clebsch-seed-certificate.json` specifies the basis as 48-bit integers, 73 original row and column indices of Q, and the rows of an inverse to the resulting 73-square minor. Bit zero is coordinate zero throughout.

Kernel membership, basis independence, the minor entries, and the identity between that minor and its supplied inverse are finite binary identities; `tools/check_clebsch_seed.py` reconstructs the graph and checks each of them using only integer bit arithmetic. In particular the minor has determinant one, so $\operatorname{rank} Q \geq 73$. The 47-dimensional cut space is contained in $\ker Q$ by Lemma 9.1; hence $\operatorname{rank} Q \leq 120 - 47 = 73$. Equality identifies its kernel with the cuts and proves rigidity. The explicit certificate is part of this finite proof; its Python verification is not a Lean formalization or a numerical rank estimate. □

Corollary B.4 (Towers above the rigid seed)**.** Not formalized; a complete proof is provided.

Every connected binary tower above F has adjacency-code dimension at most $N/2+2$, with the same equality/rigidity versus permanent at-most-half dichotomy as Theorem 9.2. Equality is attained at every level. The bound also holds for connected regular 2-group covers of F.

Proof (not formalized). Initially the excess is $26-48/2 = 2$ and the graph is rigid. Apply Lemma 9.1 and the induction in Theorem 9.2. The seed contains the ordinary Clebsch kernel pulled back to all sheets, so endpoint evaluation is onto on every seed edge. It persists in binary lifts, making every single-edge signing have loss two and proving sharpness. The subgroup-chain argument gives the regular-cover assertion. □

This also handles a conditional noncentral extension: suppose $1 \to P \to Q \to S_3 \to 1$ has 2-group kernel and a connected regular Q cover has this *actual* three-point quotient F. The preimage in Q of a point stabilizer has order $2|P|$; its right action is a regular deck action over F. The preceding corollary applies. Merely possessing an S_3 quotient does not identify that quotient cover with F.

These results extend the available finite obstructions to specified noncommuting actions. They do not address every permutation cover, arbitrary subsequent bases, or unrestricted recovery codes. The counterexample in Proposition A.3 explains why replacing the remaining permutation problem by arbitrary reciprocal blocks loses a potentially essential restriction.

C Index of Lean declarations

Declarations are relative to `CoverStorage`; their axiom checks are generated together with this index (Appendix D).

`CoverStorage.HomogeneousInflation` `finrank_kernel_inflated`, `inflatedCheck`, `kernelEquiv`, `rate_cross_product`, `reindex_inflated`, `sheetCheck`, `sheetReindex`.

`CoverStorage.LocalGeneration` `kernel_range_dimension_bound`, `quotient_dimension_bound`, `split_generator_dimension_bound`.

D Proof status and verification

The Lean-verified results are exact homogenization (Proposition 8.1) and the dimension interfaces (Lemma 10.1). The shared `LeanMachine/HomogeneousInflation.lean` module constructs the two checks and their coordinate permutation, proves the kernel equivalence, and applies Mathlib's finite-product dimension formula. `LeanMachine/KernelRangeBound.lean` reuses Mathlib's dimension monotonicity, rank-nullity, quotient dimension, and span closure for the dimension, quotient-deficit,

and split-generator bounds. The paper modules expose these results without duplicating their proofs. The English outlines are guides; the Lean declarations are authoritative.

The remaining graph, finite and asymptotic cover, nonlinear-source, and group-word results have complete written proofs. In particular, the local-generation theorem is not described as formalized merely because its final dimension inference is checked. The native hypotheses, free-group separation, imported Cohn and Jaikin-Zapirain–Souza theorems, parameter uniformization, coefficient filtrations, weighted determinant argument, and full-matrix-algebra inflation example are not formalized in this manuscript. Nor is the written product-subcode corollary part of the formal kernel equivalence. The source-coset argument and central-layer group-word specialization are also written proofs, not additional claims about the scope of that interface. The README's formalization todos record the remaining dependencies and existing library material to reuse. No floating-point estimate or experimental observation is used as a proof premise.

The finite Clebsch lift, reciprocal-block, and small-representation arguments are complete written proofs. Their algebraic base signing rigidity proof does not depend on a finite rank computation. The separate 48-vertex seed uses the exact witness in `tools/clebsch-seed-certificate.json`: its literal kernel basis, original signing-map minor, and explicit inverse are verified by `python3 tools/check_clebsch_seed.py`. This finite identity certificate is part of the written proof and is not described as kernel-checked Lean code. It uses no floating-point arithmetic or solver.

Lean and Mathlib are pinned to v4.34.1. The paper requires the sibling shared package through a relative path. Clone the whole repository with its directories intact. From the paper directory, `python3 tools/lean_index.py` generates the declaration index and `lean/PaperAxioms.lean` from the same annotations. In `lean/`, run `lake exe cache get`, `lake build`, and `lake env lean PaperAxioms.lean`. The cited declarations use at most `propext`, `Classical.choice`, and `Quot.sound`. In `paper/`, run `latexmk -pdf main.tex`.

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